

Computer Vision Homework 1

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PART 1. Write a program to do the following requirement.

For part 1 I am using python 3.10 to implement all the image processing requirements.

Below are the results and brief explanations

(a) upside-down lena.bmp



Explanation :

Using a nested loop to go through and extract all the pixel row indexes, and then pass the row values in reverse. So, the top row gets the pixels from the bottom row, the second row gets the second from the bottom, and so on.

(b) right-side-left lena.bmp



Explanation :

Using a nested loop to go through and extract all the pixel column indexes, and then pass the column values in reverse. So, the far-left column gets the pixels from the far-right column, the second column gets the second from the right, and so on.

(c) diagonally flip lena.bmp



Explanation :

Using a nested loop to go through the image and switch all the indexes of columns and rows. pixel's row index becomes the new column index, and the pixel's column index becomes the new row index.

PART 2. Write a program or use software to do the following requirement.

For part 2 I am using online web-based image processing software Photopea to implement all the image processing requirements. (<https://www.photopea.com/>)

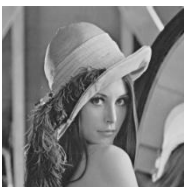
(d) rotate lena.bmp 45 degrees clockwise



Explanation :

Using free transform tool and then insert the rotation to 45 degrees clockwise.

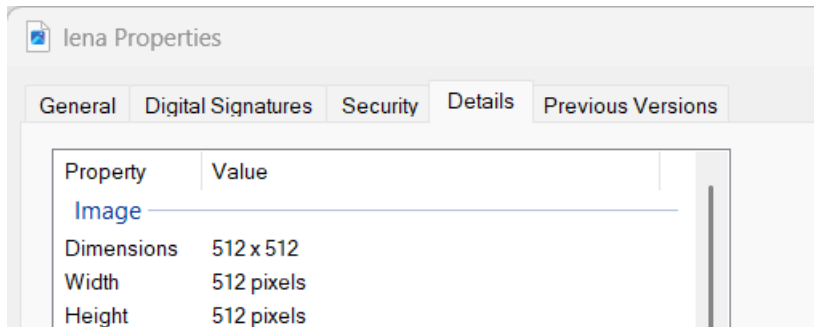
(e) shrink lena.bmp in half



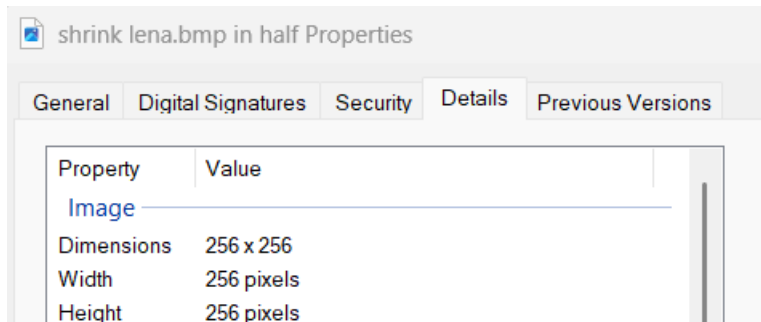
Explanation :

Using Image > Image Size feature, change the unit to percent (%) and then proceed to input the size to be 50%.

Attachments below are the comparison between the original and shrunk image dimension.
Before/Original :



After :



(f) binarize lena.bmp at 128



Explanation :

Using Image -> Adjustment -> Threshold tool to adjust the threshold value to be 128